PROJECT SYNOPSIS

AUTOSAR HLD Intelligence & Traceability Assistant

AI-assisted analysis, search and validation of automotive high-level design documents

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01 Abstract

High-level design (HLD) documents describe system components, interfaces, ports, signals, dependencies and functional flows. Finding a specific design fact or tracing it to the correct revision can require repeated manual review. The AUTOSAR HLD Intelligence & Traceability Assistant is a prototype intended to make this information easier to search, inspect and compare. It combines PDF content extraction, structure-aware chunking, metadata-preserving vector retrieval, retrieval-augmented generation (RAG), source citations, deterministic architecture checks and a human-review workflow. The documented configuration uses a Streamlit interface, FastAPI service layer, Ollama with Qwen2.5:3B for local language-model inference, BAAI/bge-small-en-v1.5 embeddings, FAISS vector search, SQLite for local workflow/governance records and Docker Compose for deployment. The current demonstration uses synthetic versioned HLD fixtures; it is not evidence of performance on real OEM documents or formal automotive standards compliance.

02 Problem Statement

Engineers reviewing lengthy HLD documents may need to locate interface definitions, compare changes between revisions, identify dependencies and verify whether a generated summary is supported by a precise source passage. Manual inspection can be time-consuming, and answers without reliable document/version/page provenance are difficult to audit. The project addresses this problem by coupling natural-language access with retrieved evidence and explicit validation/review signals.

03 Aim and Objectives

- Create a searchable knowledge base from approved HLD documents while preserving document, version, page, section and chunk metadata.
- Answer natural-language questions using retrieved passages and display citations that reviewers can check against source material.
- Support architecture validation, inconsistency checks and revision comparison using implemented deterministic Python logic.
- Evaluate response correctness, retrieval relevance, citation validity, grounding, cross-version attribution and latency.
- Retain human oversight and communicate uncertainty, known limitations and cases requiring engineering review.

04 Intended Users and Scope

Intended users include system/AUTOSAR architects, software and integration engineers, test engineers, and engineering reviewers. The prototype scope includes PDF ingestion and extraction, OCR fallback where configured, architecture-aware chunking, embeddings and vector search, evidence-grounded Q&A, citation/grounding checks, version comparison, architecture validation and review-oriented output. It is an engineering aid rather than an automatic approval system.

PROPOSED SOLUTION & EVALUATION

05 Methodology and Workflow

Ingest	Load an approved HLD PDF and retain document identity and version information.
Extract	Extract text, page content, headings and tables where possible; use OCR fallback for scanned pages when available and configured.
Prepare knowledge base	Split the content into structure-aware chunks with source metadata, then create BGE embeddings and store vectors in FAISS.
Retrieve and generate	Embed the user query, retrieve a configured top_k evidence set, and pass the evidence to the local Ollama/Qwen model through the RAG prompt.
Check and present	Check claim support, citation metadata, version alignment and contradictions where implemented; show the answer, evidence and review indicators.
Review and compare	Let an engineer inspect cited source sections and use the implemented comparison/validation logic to investigate possible changes or inconsistencies.

06 Technology Stack

Interface	Streamlit
API / orchestration	FastAPI and Uvicorn
PDF extraction	PyMuPDF; OCR fallback where configured
Embeddings / retrieval	BAAI/bge-small-en-v1.5 and FAISS
Language model	Ollama with Qwen2.5:3B
Persistence / deployment	SQLite for local workflow/governance; Docker Compose

07 Evaluation Plan and Current Test Context

Evaluation should use fixed questions with expected answers and source versions. Compare Baseline and Hybrid using the same top_k, dataset/index, embedding model, LLM, prompts and generation settings. Measure answer correctness, retrieval relevance, citation validity, grounding, version fidelity, latency and review flags; retain failed as well as successful runs.

In the synthetic VehicleSpeed regression case, v1.0 is float32 and v2.0 is float64. Two recent reports returned the correct distinction with valid citations; earlier runs contained incorrect answers or citation/evidence issues. These observations do not establish a stable success rate. Repeat the test on the exact final build.

08 Responsible AI, Security and Human Oversight

- Grounding: expose source citations and use claim-level support/contradiction checks where implemented; treat scores as signals, not guarantees of truth.
- Privacy and security: use only documents approved for the environment; keep secrets out of source control, reports, screenshots and logs; share only placeholder environment settings.
- Human oversight: engineers must verify cited evidence and retain responsibility for conclusions and architecture decisions.
- Scope of claims: the prototype is not claimed to be certified, safety-rated, or formally compliant with AUTOSAR or ISO 26262.

09 Expected Outcomes and Limitations

Expected outcomes include searchable HLD content, citation-backed answers, traceable evidence, and tools to surface candidate architecture changes. Limitations include CPU inference latency, complex PDF/table extraction, and cross-version errors in earlier runs. Broader approved-document tests and a repeatable Baseline-versus-Hybrid benchmark are required before claiming stronger reliability.

Keywords: AUTOSAR HLD, retrieval-augmented generation, vector search, FAISS, Qwen2.5, traceability, architecture validation, citation grounding, human oversight.